

Management & Staff EHS Induction

KAZ Power Plant Upgrade Project

Khor Al-Zubair Gas Turbine Power Plant · Basrah, Iraq

Client: Ministry of Electricity (MoE) & Basrah Gas Company (BGC) · Principal Contractor: Siemens Energy

Subcontractor: Al-Mial Company

Prepared by: Ahmed Al-Mansoury | EHS Manager – Siemens

01

Introduction and Objectives

How this session runs, what you will learn and what is recorded.

Session format · Learning objectives · Validity · Assessment

How This Session Runs

Emergency instructions for the training room

- If an emergency occurs during this induction: stop, follow the exit signs and go to the nearest assembly point.
- Alarm signals: continuous siren = evacuate · intermittent = stand by · all-clear = return to work.
- Mobile phones on silent — keep your phone available for emergencies.
- Smoking is prohibited except in the designated smoking area, and never in red / restricted zones.
- Duration: approx. 75 minutes, including the knowledge check and questions.

Emergency numbers are on slide 47 — know them before you leave this room.

Contents

Six chapters — about 75 minutes, including the knowledge check

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Learning Objectives

At the end of this induction you will be able to:

- State the Siemens EHS target and the Zero Harm policy, and explain what it means for your work.
- Describe your personal responsibilities and the authority you hold to stop unsafe work.
- Apply the general site rules and select the correct PPE for your task.
- Read a risk assessment, complete an LMRA and know when an SJA is required.
- Explain how the Permit to Work and LOTO systems operate on this site.
- Recognise the electrical, height, lifting, excavation, confined-space and hot-work hazards of KAZ.
- Respond correctly to a fire, an emergency, an injury, a spill or a security event.
- Report incidents, near misses and unsafe conditions within the required timeframes.

Validity, Assessment and Records

This induction is part of your site authorisation

- Validity: 12 months from the date of attendance; renewal is required on expiry.
- Re-induction is required after a serious incident or when the hazards of your work change.
- Knowledge check: 10 questions — pass mark 80%. Below 80% you are retrained before site access.
- Your attendance and signature are recorded in the site training register.
- You must also complete Personal Safety Instruction (PSI) and, for high-risk activities, the Siemens Energy Safe Start session.
- The induction card must be carried and shown on request — it does not replace the Permit to Work.
- Minimum age for site access: 18 years.

02

EHS Leadership and Responsibilities

The Siemens EHS target and who is accountable for what on this site.

Zero Harm · EHS Target · Leadership · Everyone · Contractors

Siemens EHS Target — Our Vision

Derived from the KAZ Project EHS Plan (Section 5)

- Our standard for EHS is excellence — it must exceed customer expectations.
- The project will promote a Zero-Harm culture that prevents damage to people, the environment and equipment.
- Project target: zero harm — measured as Zero H1 = ZERO (no lost-time injuries).
- Where EHS, progress and finances appear to conflict, EHS always prevails.
- Health and safety are conditions of every activity — never a trade-off against schedule.

Zero Harm is not a slogan: it is the way we plan, verify and behave on this site.



Zero Harm — we take care of each other

The Zero Harm Framework

Principles · Behaviours · Essentials

■ Principles:

- Zero incidents is achievable.
- Health and safety — no compromises.
- We take care of each other.

■ Behaviours:

- Take the time to assess risk — plan and organise every job before you start.
- Talk about safety before the work and again during the shift.
- If a task cannot be done safely, or conditions change — stop and re-assess.
- Follow the rules and procedures, including site-specific rules and PPE.
- Intervene when you see risk — never walk by.

■ Essentials: Stop principle · safe systems of work · competency · risk assessment · permit to work · PPE · emergency preparedness.

Leadership Responsibilities

For managers, engineers and supervisors

- Own EHS in your area — you are accountable for the people and the tasks you control.
 - Provide the resources: competent people, correct tools, valid PPE, adequate time.
 - Verify compliance — check permits, certificates, controls and housekeeping every day.
 - Lead visibly: safety walks, toolbox talks, safety stand-downs and recognition of good practice.
-
- Make EHS the first agenda item in every formal meeting and record it in the minutes.
 - Stop any work you consider unsafe, and support anyone who stops work.
 - Ensure incidents and near misses are reported and investigated — and lessons are applied.

Everyone's Responsibilities

Applies to employees, contractors and visitors

- Take care of your own health and safety and that of everyone around you.
- Comply with site rules, permits, method statements and instructions.
- Report hazards, incidents, near misses, unsafe acts and unsafe conditions.
- Use and maintain PPE correctly; never work with defective equipment.
- Attend inductions, PSI, toolbox talks and task-specific training.
- Never work under the influence of alcohol or drugs — random testing is enforced.
- Protect the environment: prevent spills, segregate waste, conserve energy and water.
- Participate in risk assessments for your own tasks — you know the job best.

Contractor & Subcontractor Obligations

How we work together

- Each contractor works to its own EHS system, enriched with Siemens Energy, Client and project requirements.
- Every contractor appoints a competent Contractor EHS Representative.
- Contractors provide: risk assessments, method statements, permits, competency certificates and equipment certificates.
- Monthly man-hours and incident data are submitted to the EHS Department.
- Siemens Energy may stop work, suspend, exclude or ban individuals for EHS violations — at the contractor's cost.
- Where local legal requirements differ from Siemens Energy requirements, the stricter requirement applies.

03

The Project, the Site and Daily Rules

Where you are working, how the site is organised and the rules that always apply.

Project scope · Site layout · Base camp · EHS organisation · General rules · Stop Work Authority · Risk assessment · Daily routine

The Project

KAZ Power Plant Upgrade Project

- Modernisation and expansion of the existing 400/132/11 kV Air Insulated Switchgear (AIS) facilities.
- New 400 kV and 132 kV bays: circuit breakers, disconnectors, earthing switches, current and voltage transformers, surge arresters and busbar extensions.
- Civil works: foundations, earthing grid enhancement, steel structures, cable trenches and ducts.
- Electrical works: control, protection, metering, telecom and SCADA/DAS interfacing.
- Activities: installation, testing, commissioning and trial operation.
- Duration: 13.5 months. Client: MoE & BGC. Principal Contractor: Siemens Energy.

Critical context: the works are executed inside a LIVE power plant — coordination with MoE Operations is mandatory for every activity.

Site Layout and Emergency Facilities

Know your location before you start work

- Your work area, mobilisation and laydown area, warehouses, camp and offices — shown on the site plot plan.
- Emergency layer you must know:
 - Assembly points (Muster points) for your area.
 - First-aid points, site clinic and the route to the nearest hospital.
 - Fire points, extinguishers and eyewash stations.
 - Emergency exits and the route for fire, ambulance and emergency vehicles.
- Red zones: high-voltage areas where access is restricted and controlled.
- The plot plan is posted at the site entrance, offices and assembly points — check it on your first day.

Base Camp and Site Offices

The main office, the camp and the facilities you will use

- **Main office block** — project management, EHS office, document control, meeting rooms and the site clinic.
- **Base camp** — accommodation, canteen, prayer room, laundry and recreation areas. Camp rules apply at all times.
- **Camp and office safety:** keep escape routes and extinguishers clear; no cooking, no smoking and no unauthorised electrical appliances inside rooms or offices.
- **Utilities** — potable water points, generators and UPS areas, waste collection points and the sewage / grey-water system.
- **Housekeeping** — bins are provided in every block; report damage, leaks, pests or a faulty air conditioner to the camp supervisor immediately.
- **Transport** — only approved site buses and drivers; report to the camp marshal before boarding and never leave site without authorisation.
- **Emergency at the camp** — the camp assembly point is marked, and the clinic, first aiders and security gate are manned at all times.



INSERT SITE PHOTOGRAPH

Right-click the frame in PowerPoint
then choose Change Picture

Base camp and main office — insert site
photograph

Project EHS Organisation

Who is responsible for what

- Project Director / Site Manager — overall responsibility for EHS performance of the project.
- Siemens Energy EHS Manager in Projects (EHSMIP) — governance, audits, verification and stop-work authority.
- Contractor EHS Manager and EHS Officers — daily supervision, permits, inspections and training.
- Area Authority / MoE Operations — authorises work in live operational areas.
- Authorized Electrical Person (AEP) / Isolation Authority — energy isolation, LOTO and energisation.
- First aiders, fire wardens and gas testers — appointed and posted in every area.

Rule: No EHS supervision = No work.

General Site Rules (1)

Applies from the moment you enter the gate

- Complete site induction and PSI before starting any work — and carry your induction card.
- No work without a valid Permit to Work; no work without EHS supervision.
- Complete an LMRA before every task and attend the daily toolbox talk.
- Read and understand the risk assessment and method statement for your task — ask if anything is unclear.
- Wear the mandatory minimum PPE at all times (slide 38).
- Stay within your authorised area — never enter red or restricted zones without authorisation.
- Keep your workplace clean and tidy — 'Clean everything, always.'
- Report every near miss, unsafe act, unsafe condition and incident.

General Site Rules (2)

Working hours, competency and behaviour

- Working hours: 06:00–15:00, six days per week; subcontractor hours must be approved by Siemens Energy.
- Maximum 11 hours daytime / 7.5 hours night work, excluding breaks; one rest day in every seven. No fatigue working or driving.
- Only competent, certified persons operate plant, equipment and electrical systems.
- No horseplay, no sleeping on site, no unauthorised removal of materials (Material Gate Pass required).
- Zero tolerance: alcohol, drugs, weapons, fighting, harassment and theft.
- Photography and video recording on site are prohibited without written authorisation from the Site EHS/Security Manager.
- Report to the site medic any illness, injury or medication that could affect your work.

Stop Work Authority

You are empowered — and expected — to use it

- Anyone — employee, contractor or visitor — may stop any task where there is imminent danger, a rule violation, a missing control or a change in conditions.
- How to apply it:
 - 1. Stop the activity.
 - 2. Intervene with the people involved — calmly and respectfully.
 - 3. Escalate to the supervisor or area authority.
 - 4. Verify and implement the required controls.
 - 5. Restart only when authorised.
 - 6. Record the event and give positive feedback.
- You will not be punished, prosecuted or harassed for stopping work — this is a Siemens Energy commitment.

Risk Management

Risk Assessment · SJA · LMRA · Management of Change

- Risk assessment (RA): prepared before work starts and communicated to the team. Read it, follow the controls, never work outside its scope.
- Safe Job Analysis (SJA): required for high-risk or non-routine tasks; prepared with the crew before work starts.
- Last Minute Risk Assessment (LMRA): a short check before every task — What can hurt me? What has changed? Are the controls in place?
- Hierarchy of controls: eliminate · substitute · engineering controls · administrative controls · PPE (PPE is the last line of defence).
- Management of Change: if the method, equipment, people or conditions change, stop work and revise the assessment.
- After any incident or high-potential near miss, the risk assessment is reviewed and updated.

The Daily Safety Routine

Start of shift · during work · end of shift

- Start of shift: toolbox talk → LMRA → permit verification → PPE and equipment pre-use checks.
- During work: supervision present, barriers and signage in place, housekeeping maintained, no lone working on high-risk tasks.
- End of shift: close and endorse permits, restore isolations per the LOTO register, secure the area.
- Shift handover for isolated work: joint re-verification of permit conditions, isolation status and zero-energy state, signed by all parties.
- Any change of plan during the shift is briefed to the crew and, where required, re-approved.

04

Permit to Work and Electrical Safety

The control system for hazardous work and the rules that keep the plant safe.

Permit to Work · Lock-Out / Tag-Out · Electrical rules · Arc flash · Live plant

Permit to Work (1)

System, roles and principles

- No permit = no work. The PTW system is administered and controlled by Siemens Energy.
- For work on live or operational plant and client-controlled areas, the MoE/BGC permit system and its Area/Isolation Authority take precedence.
- Roles: Permit Issuer (SE) · Permit Holder (contractor supervisor) · Area Authority (operations) · Isolation Authority / AEP · EHS Reviewer · Authorized Gas Tester.
- Every permit requires: approved JSA / method statement, controls verified on site, briefed working party, valid certificates and inspections.
- EHS reviews and audits permits; technical approval belongs to the Area Authority; isolation authority belongs to the AEP.
- Never work outside the scope, time or conditions stated on the permit.



No permit — no work

Permit to Work (2)

Permit types and validity on this project

Permit	Typical validity
General Cold Work Permit	7 days with daily endorsement
Electrical Cold Work Permit	7 days with daily endorsement
Electrical Hot Work & Commissioning Permit	Daily / each shift
Energy Isolation (LOTO) Permit	Daily / each shift
Hot Work Permit (welding, cutting, grinding)	Daily / each shift
Excavation & Trenching Permit (over 30 cm)	7 days with daily endorsement
Lifting Permit (all lifts; engineered plan for critical lifts)	Daily / each shift
Work at Height Permit (1.8 m and above)	Daily / each shift
Confined Space Entry Permit	Daily / each shift

Lock-Out / Tag-Out and Energy Isolation

Control every energy source

- Cover all energy sources: electrical, mechanical, pneumatic, hydraulic, thermal and stored energy.
- Sequence: notify → isolate → lock and tag → release stored energy → verify zero energy → work.
- Every worker applies their own personal lock; keys are controlled in the lockbox by the AEP.
- PTW validity and LOTO validity are managed separately — permit expiry never releases an isolation.
- Only the AEP de-isolates, through the controlled de-isolation process and the key register.
- Never remove another person's lock or tag — this is a dismissible offence.
- Test before touch: prove the tester, prove dead, and prove the tester again.

Electrical Safety Rules

The golden rule of this project

- Treat every electrical circuit as live until proven dead.
- Isolate the source and apply LOTO before starting any work.
- Only an Authorized Electrical Person may isolate, de-isolate or energise.
- Verify absence of voltage with an approved, calibrated tester — prove the tester before and after.
- Use insulated tools rated for the voltage; inspect them before use.
- Maintain safe approach distances at all times; respect barriers, interlocks and signage.
- Never work on live equipment unless formally authorised and only under an Electrical Hot Work / Commissioning Permit.

GOLDEN RULE: ISOLATE — LOCK OUT — TAG OUT — TEST BEFORE TOUCH.



Arc-rated PPE and approach boundary

Arc Flash and Safe Approach Distances

Boundaries must be approved for this site — do not use generic values

Nominal voltage	Minimum approach distance	Arc flash boundary	Notes
11 kV	per approved project table	per arc flash study	Limited / restricted approach per NFPA 70E and client rules
132 kV	per approved project table	per arc flash study	Client operational rules apply
400 kV	per approved project table	per arc flash study	Highest risk on this site — client operational rules apply
Work within 30 m of HV system	Agreement with plant owner + physical demarcation	—	Mandatory before any activity

Working in a Live Power Plant

Coordination with MoE Operations

- Work near or on live systems is governed by the client's operational rules in addition to ours.
- A nominated Operations Manager is responsible for the installation; a Switching Supervisor controls switching.
- No work starts without: agreed method, permit, isolation and the operational authority's confirmation that the plant condition is safe.
- Maintain physical demarcation between live and work areas; never breach a barrier or open a panel without authorisation.
- Stop all work during thunderstorms — cranes, lifts, scaffolding and installation work are suspended (lightning and induced voltage).
- Report any contact with a conductor, any flash and any damaged insulation immediately.
- Never touch an electric-shock casualty before the supply is isolated.

05

High-Risk Activities and Occupational Health

The activities that cause the most serious injuries and how each one is controlled.

Work at height · Lifting · Excavation · Confined space · Hot work · Tools · Manual handling · PPE · Heat stress · First aid

Work at Height

Fall protection from 1.8 m

- Fall protection is required from 1.8 m, or wherever a fall hazard exists.
- Use approved scaffolding, work platforms or MEWPs — with guardrails and toe boards.
- Harness with double lanyard and shock absorber, 100% tie-off to a certified anchor point (above 22 kN).
- Scaffolding: erected and altered only by certified scaffolders; green tag valid; inspected before each shift and after any alteration. Never work on a red-tagged scaffold.
- Ladders: short-duration work only, secured, three points of contact, no loads carried.
- Barricade the area below, tether tools, protect openings and edges, and never work at several levels without protection.



100% tie-off to certified anchor

Lifting Operations

Every lift is planned and permitted

- All lifts require a permit; critical lifts require a detailed engineered lifting plan.
- Critical lifts include: loads above 75% of rated capacity, multi-crane lifts, lifts over live plant, and non-routine rigging.
- Only certified cranes and lifting tackle (third-party certificates), competent operators, riggers and banksmen.
- Outriggers fully extended on timber pads on firm, verified ground.
- Exclusion zone barricaded — nobody under the load, no side pulls, no shock loading.
- No lifting when wind exceeds 32 km/h or in poor visibility.
- Taglines for load control; the banksman is the only person who signals the operator.



Exclusion zone · taglines · banksman

Excavation and Trenching

Permit required beyond 30 cm

- A permit is required for any ground penetration deeper than 30 cm.
- Obtain the signed underground utility clearance before digging; hand-dig trial pits near services and the earthing grid.
- Protective systems — shoring, benching or sloping — are mandatory for excavations deeper than 1.2 m.
- Spoil heaps set back at least 1.0 m from the edge; materials and plant kept away from the crest.
- Barricade, sign and light all excavations; access ladders every 7.5 m for emergency egress.
- Inspect after rain or any ground movement; never enter an unprotected trench.
- A competent banksman controls plant movement around the excavation.

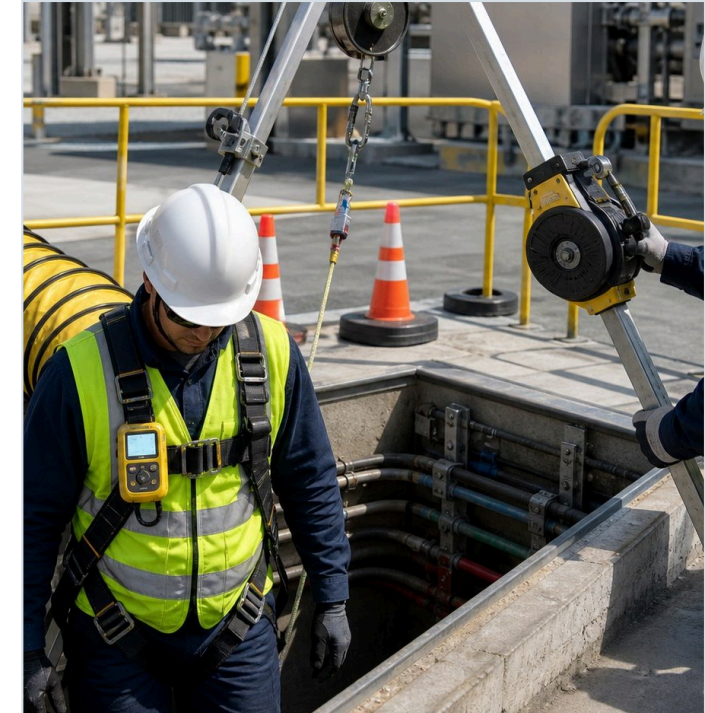


Shoring · barriers · access ladder

Confined Space Entry

Permit · gas test · attendant · rescue plan

- Entry only with a Confined Space Entry Permit, trained entrants and attendants, and a rescue plan in place.
- Atmospheric testing before entry and continuously during the work: oxygen, LEL, H₂S and CO, by an Authorized Gas Tester.
- Ventilation and isolation (LOTO) in place before entry.
- The attendant remains at the entry at all times and maintains communication with the entrant.
- Never enter to rescue without breathing apparatus and the rescue team — most confined-space fatalities are would-be rescuers.
- Classification as a confined space is based on risk assessment of the actual conditions.

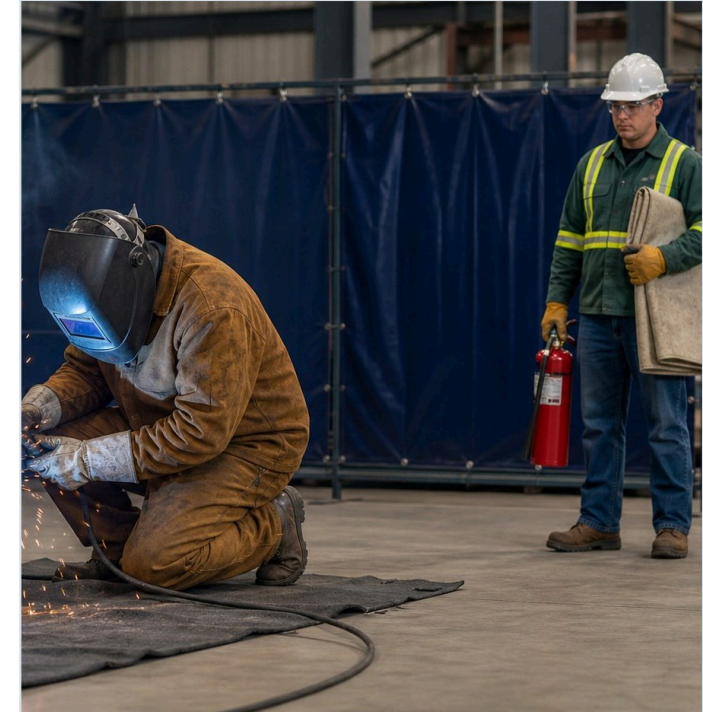


Gas monitoring · attendant · tripod

Hot Work and Fire Watch

Sparks and flames are controlled, permitted activities

- Hot Work Permit required for welding, cutting, grinding, soldering, thermal spraying or any activity producing sparks, flames or heat above 100 °C.
- Certified welder or cutter; PPE and screens to protect others.
- Fire blanket containment and at least 10 m clearance of combustible materials.
- A dedicated Fire Watch is present during the work and for 30 minutes after it ends.
- Minimum two extinguishers and a fire blanket at the work site.
- Gas cylinders secured upright, caps fitted, oxygen and fuel gas stored separately.
- Gas test required near fuel, chemical or potentially flammable areas — never hot work in a flammable atmosphere.



Fire watch · containment · 10 m clearance

Hand and Power Tools

The right tool, inspected, used correctly

- Use the right tool for the job; inspect before use; remove defective tools from service immediately.
- Portable electrical tools: 30 mA RCD protection, undamaged armoured cables, earthing verified, no taped joints.
- Angle grinders: guard in place, correct disc for the material, never over-speed a disc.
- Wear a full face shield (EN 166) plus safety glasses and gloves for grinding and cutting.
- Secure the workpiece; never cut towards your body; keep hands clear of the cutting line.
- No loose clothing, no jewellery, no rings; tie back long hair.
- Never use compressed air to clean clothing or skin.

Manual Handling and Ergonomics

Plan the lift, not just the job

- Assess the load: weight, size, grip, centre of gravity, distance and destination.
- Use mechanical aids or ask for help — team lifts need a leader and clear commands.
- Lift technique: bend the knees, keep the load close, no twisting while lifting.
- Plan breaks and vary tasks to reduce repetition and strain.
- Report pain early — discomfort today becomes an injury tomorrow.
- Consider the environment: heat, uneven ground and confined spaces all increase the risk.

Personal Protective Equipment

Mandatory minimum on site — additional PPE per task risk assessment

PPE item	Requirement on this site
Safety helmet	Mandatory — with chin strap; replaced after impact
Flame-retardant coverall	Mandatory — visibility class 1 (high-visibility)
Safety footwear	Mandatory — S3 / S1P, toe protection and puncture resistance
Safety glasses	Mandatory — ANSI/EN certified
Protective gloves	Mandatory — task-specific (electrical, chemical, cut-resistant)
Hearing protection	Mandatory above 85 dB(A) or where indicated
Arc-rated PPE	For electrical work — Category 2/4, face shield, insulated gloves
Full-body harness	Working at height — double lanyard with shock absorber
Respiratory protection	Where dust or vapour is identified — face-fit tested (e.g. FFP3)
High-visibility vest	Mandatory in areas with mobile plant and vehicle movement

Heat Stress Management

Our most predictable environmental hazard

- Heat stress is managed by WBGT monitoring — monitoring starts when ambient temperature exceeds 40 °C.
- Published work / rest cycles apply for light, moderate and heavy work; follow them.
- Acclimatisation: new arrivals and anyone returning from leave start with reduced workload and increased rest.
- Drink water regularly — about 1.0 to 1.2 litres per hour for light work in high heat. Use oral rehydration salts.
- Shaded, ventilated rest areas and chilled drinking water are provided — use them.
- Lone working is prohibited on high-risk tasks — use the buddy system.
- Warning signs: headache, dizziness, cramps, nausea, confusion, hot dry skin, fainting. Heat stroke is a medical emergency — cool the casualty and call the medic immediately.



Shade · hydration · work-rest cycles

First Aid and Medical Emergency

Know your nearest help

- Know: your nearest first-aid point, the site medic, the emergency number and the route to the nearest hospital.
- Do not move a casualty unless there is immediate danger to life.
- Electric shock: do not touch the casualty before the supply is isolated.
- Keep the airway open; start CPR / AED if the casualty is not breathing and you are trained.
- Preserve the incident scene; do not disturb equipment involved in the incident.
- Meet the ambulance at the main gate and escort it to the casualty.
- Report every injury — including first-aid cases — to the EHS Department and the site medic.

Fitness for work: a valid medical certificate is required; report any condition or medication that affects your ability to work safely.

06

Environment, Emergency and Security

Protecting the environment, responding to emergencies, security and the consequences of breaking the rules.

Environment · Waste · Fire · Emergency · Incident reporting · Security · Driving · Penalties

Environmental Protection

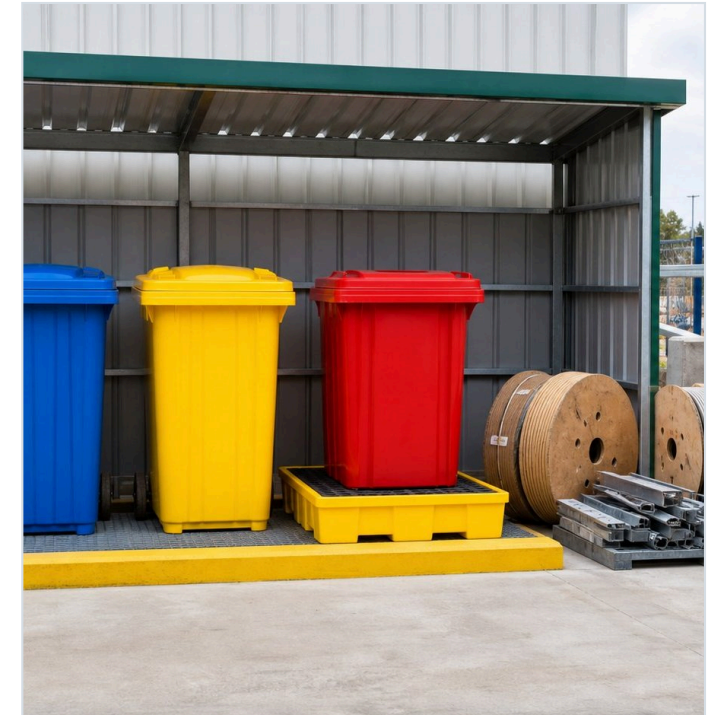
Prevent pollution — it is part of every task

- No spill of oil, fuel, chemical or concrete washwater to soil, drains or watercourses.
- Use drip trays, secondary containment, funnels and spill kits; keep spill kits stocked and accessible.
- Dust suppression on unpaved roads and during cutting, grinding or excavation.
- Control noise and vibration through plant selection and work methods; respect site noise limits.
- Chemicals: no substance on site without a Safety Data Sheet (SDS) and registration in the electronic substance register.
- Asbestos and PCBs are prohibited on site — report any suspected material immediately.
- Protect flora, fauna and heritage: no uncontrolled clearing, no hunting, no littering.
- Report environmental incidents (spills, leaks, uncontrolled emissions) immediately — they are EHS incidents.

Waste Management

Segregate at source — every day

- Segregate waste: general · recyclable (metal, cable, paper, plastic) · hazardous · construction and demolition waste.
- Hazardous waste (oils, filters, batteries, chemicals, e-waste) is stored in labelled, covered, bunded areas.
- Hazardous waste leaves site only with MOTAT records and a licensed transporter.
- Never burn waste, never mix hazardous with general waste, never dump into trenches or open ground.
- Material and waste removal requires a Material Gate Pass.
- Keep waste areas tidy, marked and away from drains; report overflowing or damaged bins.



Segregated bins · bunded store

Fire Safety and Firefighting

Classes, extinguishers and the PASS method

- Class A — wood, paper, cloth, rubber, plastics → water or foam (DCP is acceptable but less effective).
- Class B — gasoline, kerosene, paint, propane → foam, DCP, CO2.
- Class C — energised electrical equipment → CO2 or clean agent, plus de-energise the supply. Never use water.
- Class D — metals such as magnesium, titanium and sodium → special metal powder (graphite or sodium chloride). Never water or standard DCP.
- Class K / F — cooking oils and fats → wet chemical (camp kitchen).
- PASS method: Pull the pin · Aim at the base of the fire · Squeeze the handle · Sweep side to side.
- For CO2 extinguishers hold the insulated horn, never the metal parts.
- Only fight a small fire if you are trained and have a clear escape route. Otherwise raise the alarm and evacuate. Always alarm first.



Aim at the base · keep an escape route

Emergency Response and Evacuation

Alarm · assembly · head count

- Alarm signals: continuous siren = evacuate · intermittent = stand by · all-clear = return to work.
- On evacuation: stop the job, isolate your equipment, proceed to the nearest assembly point.
- Report to your area marshal for the head count; report any missing person immediately.
- Do not use vehicles; do not block emergency routes; keep the gate clear for emergency services.
- Do not leave or return to the area until instructed by the emergency controller.
- For security incidents (threat, intrusion, civil disturbance) follow the same assembly principle and the security team's instructions.
- Fire wardens sweep their areas, including visitors and subcontractor crews.



Assembly point · roll call · clear routes

Emergency Classification and Notification

Know the level — and who to tell

Level	Description	Notification
Level 1	Minor injury or health effect, minor damage or minor environmental impact, manageable on site with site resources.	Supervisor → EHS → Site Manager (verbal immediately, written within 24 hours)
Level 2	Significant event manageable locally with support from the Company or local authorities — major injury, local damage, local impact.	EHS → SE EHSMIP → Project Director + Client (immediately)
Level 3	Major event or crisis beyond the contractor's control, with broader implications, requiring Company / corporate resources.	Project Director → SE Corporate / Regional Hub + Client (immediately)
Escalate immediately	Life-threatening injury, fatality, high-potential incident, major environmental release, security crisis — never treated as a locally managed event.	Verbal immediately · written within 24 hours · investigation report within 7 days

Emergency Contacts

Posted at every assembly point, office and camp

Contact	Number / channel
Site EHS control room / emergency line	to be inserted (24/7)
UHF radio	Channel 1
Site medic / first aiders	to be inserted
Contractor emergency numbers	0785-285-7404 · 0786-162-9706
Siemens Energy emergency number	+90 533 625 9494
Ambulance (Iraq)	122
Civil Defence / Fire (Iraq)	115
Police (Iraq)	104
Nearest hospital / medevac	to be inserted

Incident, Near Miss and Unsafe Condition Reporting

Report immediately — reporting is protected

- Report immediately: all injuries (including first aid), near misses, unsafe acts, unsafe conditions, property damage, environmental releases and security events.
- Report through your supervisor and the EHS Department; incidents are recorded in the Siemens Energy EHS reporting tool (Enablon).
- Timelines:
 - Verbal notification — immediately.
 - Written notification — within 24 hours.
 - Investigation report — within 7 days (unless an extension is agreed).
 - Corrective action status — reported weekly until closed.
- A formal investigation team is formed by the Contractor, the Company and subcontractors; root causes and actions are documented.
- Never withhold information, alter the scene or delay reporting. Honest reporting carries no blame.

Learning from Incidents

Every event makes the site safer

- Lessons Learned and safety alerts are issued after incidents and high-potential near misses.
- Each lesson triggers a risk assessment review, a toolbox talk and verification that controls are effective.
- Siemens Energy may conduct Eye on Safety reviews and its own investigations.
- If you are involved in or witness a serious incident: preserve the scene, cooperate fully and provide a written statement.
- Share your experience — it prevents the next accident.

Security and Access Control

Everyone is searched — everyone is protected

- Site and camps are protected by the Iraqi Police and site security; persons, belongings, vehicles and deliveries may be searched on entry and exit.
- Wear your Site ID / Gate Pass visibly at all times; access without it is refused.
- Visitors: pre-approval 24 hours in advance, photo ID deposited at the gate, escorted by a host at all times; no work, no plant operation, no photography.
- Report immediately: unauthorised access, theft, suspicious persons or vehicles, security breaches or threats.
- Lost or damaged ID card must be reported; a replacement fee applies.
- Security personnel are authorised to carry weapons — cooperate calmly and follow their instructions.
- Never resist or interfere with security checks; the minimum age for site access is 18 years.



Access control · vehicle search

Driving Safety and Journey Management

The road is our highest-risk journey

- Seat belts for all occupants, in every vehicle, at all times.
- Valid Iraqi driving licence for the vehicle class; only authorised drivers and vehicles.
- Site speed limit: 5 km/h. Obey traffic signs, signals and marshals.
- No mobile phone use while driving — including hands-free. No other distractions.
- Reverse parking only in designated areas; reversing requires a trained banksman.
- Daily pre-use checks and valid third-party inspection certificates for all vehicles.
- Journey management: plan the route, rest at least every two hours, no night driving without approval, no driving when tired.
- Driving is restricted in adverse weather — fog, heavy rain and dust storms.

Violations and Penalties

Consequences are applied consistently

Violation	1st offence	2nd offence	3rd offence / critical
Speeding (over 5 km/h)	Warning + 2-day vehicle ban	1-week vehicle and driver ban	Permanent site ban
PPE non-compliance	Work stoppage + written warning	Contractor fine	ID cancellation and site removal
Red zone / LOTO breach	Work stoppage + 1-week suspension	—	Permanent site expulsion
Lost or damaged ID card	Replacement fee	—	—
Positive alcohol / drugs test	Immediate permanent site ban	—	—
Unauthorised material removal	Confiscation and investigation	—	—
Unauthorised photography	Media deletion and warning	—	Permanent site ban

Knowledge Check

10 questions — pass mark 80%

- 1. What is the site speed limit, and when is a banksman required?
- 2. At what height does fall protection become mandatory?
- 3. When is a protective system required for an excavation?
- 4. Which gases are tested before confined space entry?
- 5. What is the maximum wind speed for lifting operations?
- 6. Which extinguisher would you use on an energised electrical panel fire?
- 7. What are the reporting deadlines for an incident?
- 8. What must you do before touching any electrical conductor?
- 9. What are the official emergency numbers in Iraq?
- 10. What happens if you stop work because you believe it is unsafe?

Induction Card and Acknowledgement

Before you leave this room

- Collect your numbered induction card — valid 12 months, must be carried and shown on request.
- Sign the attendance and acknowledgement register.
- The card does not replace the Permit to Work, PSI, Safe Start or task-specific training.
- Acknowledgement: 'I attended the KAZ Management & Staff EHS Induction, I understood its content, and I accept that compliance with safety rules, permits and Stop Work Authority is a condition of my access to this site.'
- Report any content gaps or unclear points to the EHS Department — this induction is reviewed regularly.



Everyone Goes Home Safely — Every Day

Zero Harm · We take care of each other

Questions? Contact the EHS Department — UHF Channel 1 / Site EHS control room.

Thank you for your attention and for the work you do safely.

Prepared by: Ahmed Al-Mansoury | EHS Manager - Siemens

No task is so urgent that it cannot be done safely.